

Use the given information to answer questions 1-3.

Quadrilateral $ADHP$ is shown with $HP = 25$, $AD = 8x + 21$, $DB = 20x - 4$ where $x = \frac{1}{2}$.

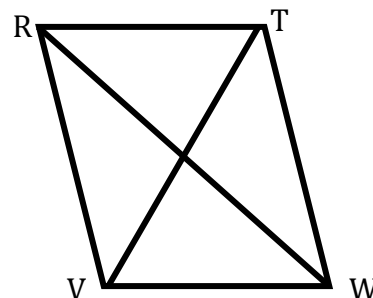


1. I can conclude ...
2. I know this because ...
3. If $ADHP$ is a parallelogram, what is the length of $\overline{PA}$?

Use the given information to answer questions 4-6.

Quadrilateral $RTWV$ is shown with diagonals $\overline{RW}$ and $\overline{TV}$ that intersect at point M where $RM = 4x$, $TM = (3y + 15)$, $WM = 84$, and $VM = 75$.

4. What value of x will make $RTWV$ a parallelogram?

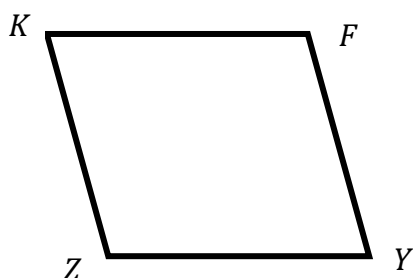


5. What value of y will make $RTWV$ a parallelogram?

6. What is the length of $\overline{VT}$?

Use the given information to answer questions 7-10.

Quadrilateral $KFYZ$ is shown where $m\angle F = (20z - 4)^\circ$, $m\angle Y = 54^\circ$, and $m\angle Z = (8t)^\circ$, where $t = 15\frac{3}{4}$.



Devonte and Briyel are discussing whether $KFYZ$ is a parallelogram. Briyel says it is, and Devonte says it is not.

7. Who is correct?

8. Which property of parallelograms was used to determine if $KFYZ$ is a parallelogram?

9. What is the value of z ?

10. What is the $m\angle K$?